SUBJECT bt405' (past papers)

CR : Tasha KHAN

VU MEDICAL ZONE BIOTECHONLOGISTS AND ZOOLOGISTS

ADMINS: Hafiza Mubeen Muhammad Nouman, Tasha Khan, Alaya sajjad

with vu medical zone biotechnologists & zoologists

Bt405

1- what are Application of self-assembled monolayers?

Applications: SAMs are extremely useful in thin-film technology since they provide a simple, versatile and relatively inexpensive way of producing coatings with specific functionalities.

- Thin coatings are used in many applications, for example to control surface wetting and adhesion properties; to provide chemical resistance or specific biocompatibility; to confer antibacterial properties; and for sensors.
- SAMs are particularly interesting for developing DNA and protein micro and nanoarrays for medical screening and for future electronic components in computer chips and other ICT developments.

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2- Nanotechnology?

3- Application of environmental metal nano particles?

Environmental applications

❑Some forms of metal nanoparticles have important environment applications.

➢Zero-valent (Fe0) iron nanoparticles are under investigation for the remediation of contaminated groundwater and soil.

➢Iron, when exposed to air, oxidises easily to rust; however, when it oxidises around contaminants such as trichloroethylene (TCE), carbon tetrachloride, dioxins, or PCBs, these organic molecules are broken down into simple, far less toxic carbon compounds.

➢Iron nanoparticles are more effective than conventional 'iron powder', which is already used to clean up industrial wastes. Iron nanoparticles are 10 to 1 000 times more reactive than commonly used iron powders.

4- Nanotechnology in diagnostic?

5- 5 ways to impact Nanotechnology on daily basis?

6- Diatoms in self healing adhesive?

3.Quantum confinement 2marks

Quantum confinement: in a nanomaterial, such as a metal, electrons are confined in space rather than free to move in the bulk of the material.

4.Difference between Intermolecular bonding and Intramolecular bonding 2marks

Intramolecular bonding (chemical interactions): these are bondings that involve chang es in the chemical structure of the molecules and include ionic, covalent and metallic bonds;

➢Intermolecular bonding (physical interaction): these are bondings that do not involve changes in the chemical structure of the molecules and include ion-ion and ion-dipole interactions; van der Waals interactions; hydrogen bonds; hydrophobic interactions; repulsive forces (such as steric repulsions).

➢Nanomaterials often arise from a number of molecules held together or large molecules that assume specific three-dimensional structures through intermolecular bonding (macromolecules)

5.Nanotechnology in Diagnostic 3marks

Application of Nanotechnology in Medicine

Diagnostic

– Imaging

– Quantum dots

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– Microscopic sampling
– Detection of airway

6.Diagnostic Applications of Nanotechnology in Medicine?

➢ Improved imaging of the human (or any) body
➢ Nanoprobes (miniature machines) can attach themselves to particles in the body (e.g., antibodies) and emit a magnetic field.
➢ Probes that aren't attached to anything don't create a detectable magnetic
➢ Nano-tracking may be able to detect tumors that are a few cells in size.
A single inhaled nanorobot reaches, deeply inspired into the lungs, enters an alveolar duct and attaches to the tissue surface.g

7.Metal nanoparticles environmental applications 3marks

ENVIRONMENT
Safer and more efficient approaches to waste management, air and water purification, reduction in pollution.
Nanofiltration to remove dissolved salts, heavy metal contaminants, soften water and treat wastewater.
Nanofibers filtrate dangerous bacteria in the air, gold nanoparticles clear volatile organic compounds.
Developed by the oil industry, MCM-41 (known also as "self-assembled monolayers on mesoporous supports," SAMMS), with pore sizes in the range of 10 — 100 nanometers, is used for the removal of ultrafine contaminants.
A nanoparticle-reinforced polymeric material can replace structural metallic components in automobiles and lead to a reduction of 1.5 billion liters of gasoline consumption over the life of one year's production of vehicles, thereby reducing carbon dioxide emissions annually by more than 5 billion kilogram

6.ways nanotechnology impacts our lives on a daily basis 5marks

CURRENT USES OF NANOTECHNOLOGY
Nano-sized TiO2 and ZnO – sunscreens
Silver nanoparticles – antibacterial properties (soaps, toothpastes, deodorants, lip products, make-up instruments, hair brushes, curling tongs, foils for electric razors, foot massagers, tooth brushes, rubber gloves, hair dryers, facial ionic steamers).
Nanosomes - improve the solubility of ingredients and add shimmer.
Nano emulsions and nanosomes -preserve active ingredients, such as vitamins and anti-oxidants.
Nano gold - healing and anti-oxident properties.
Fullerenes, bucky balls - anti-oxidant and smoothing properties in moisturisers, increased penetration into the skin , anti-aging properties.

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FOOD & PACKAGING: Natural and engineered nanomaterials in the food, additives and packaging.
New flavours and textures, less use of fat, enhanced absorption of nutrients, blockage of ingredients that contribute to elevated blood cholesterol.
Smart packaging identifying contaminated food, nanoscale sensors identifying the presence of bacteria and releasing chemicals as food spoils.
Nanomaterials preventing adhesion of the microbes to the surfaces and equipment.
Nanosilver - food contact utensils and containers.
TEXTILE AND CLOTHING: Stain resistant, waterproof, wrinkle resistant, antibacterial, radiation absorbency properties , waterproof, windproof, ultraviolet protection, breathable, grime resistant fabrics.
SPORTS & LEASURE, HOUSEHOLD: Tennis rackets, skis, snowboards, goslf balls, nanobike, baseball bats, sports clothing.
Nanoceramic- smoothness and heat resistance of household equipment (flat iron)
Nanoparticles - household appliances to kill bacterial and odors (refrigerators), shoe protection nano-sprays, etc.
COMPUTING & ELECTRONICS: Computer technologies, mobile phones, digital cameras and other high technology equipment.
Moore's law decrease in size and increase in density
Faster and smaller non-silicon-based chipsets, memory and processors, new materials for semiconductors that increase processing speeds, advanced microscopy, faster and smaller telecom switches, higher-speed transmission, new class of display using carbon nanotubes as emission device, flat-screen TVs and computer monitors.
TRANSPORT, BUILDING, CONSTRUCTION: Nanotechnologies in transport- lighter and stronger materials, more efficient fuels (cars, aeroplanes, ships)
Paints with improved adhesion and anti-mildew properties, insulation with increased insulating properties, concrete and steel with increased strength and durability, glass that acts as a fire barrier (Nanoforum), self-cleaning sheet glass based on a layer of titanium dioxide nanoscale particles (nano titanium dioxide is fixed in layers onto glass which enables sunlight to break down any dirt making it easy for the rain to wash the dirt away on ships and tankers) (Meandnano).
MEDICAL_Use
Diagnostic devices, contrast agents, analytical tools, physical therapy applications and drug delivery vehicles.
Cancer diagnosis - “molecular imaging” intended to improve diagnostic accuracy.
Targeting and localized delivery of drugs - delivering drugs specifically to cancer cells using nanoparticles that can be loaded with drugs

The US Food and Drug Administration has already approved first generation nanodrugs such as Abraxane, a nanoformulation of the anti-cancer chemotherapy paclitaxel
An anti-cancer drug approved by Health Canada containing a nanomaterial designed to find cancer cells, which attaches itself to the cells and then releases a chemical that kills them

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The drug is only released when attached to the cancer cell, the toxic side effects associated with many drugs used for chemotherapy are thereby reduced
Further advances in nanomedicine may help to reproduce or to repair damaged tissue and replace today's conventional treatments like organ transplants or artificial implants.
ENVIRONMENT
Safer and more efficient approaches to waste management, air and water purification, reduction in pollution.
Nanofiltration to remove dissolved salts, heavy metal contaminants, soften water and treat wastewater.
Nanofibers filtrate dangerous bacteria in the air, gold nanoparticles clear volatile organic compounds.
Developed by the oil industry, MCM-41 (known also as "self-assembled monolayers on mesoporous supports," SAMMS), with pore sizes in the range of 10 — 100 nanometers, is used for the removal of ultrafine contaminants.

7.Explain self healing, why diatom serves as model of biomimetic membranes? 5marks

Self-healing adhesives
➢· Diatoms are a type of algae with nanostructured amorphous silica surfaces.
➢· Diatoms are unicellular algae that have the capability to colonize any natural and manmade submerged surfaces.
➢· Diatoms are a large group of eukaryotic microalgae that have a rigid cell wall made of silica (SiO2).
➢· Diatoms are often among the first organisms to colonize underwater surfaces, because their adhesive mucilage enables the cells to attach to substrata of a very wide range of chemical composition (organic, inorganic, natural or man-made)./,,,

➢ Both diatoms and molluscs have strong underwater glues that can also resist stress and self-heal.

They serve as a biomimetic model for self-healing materials.
Self-healing adhesives
➢ These natural adhesives and their self-healing properties are due to the properties of the proteins contained in them.
➢ These proteins have sacrificial bonds that allow the molecule to be reversibly stretched by breaking and re-bonding.
➢ This sacrificial bond behaviour has been observed in many other materials, such as wool. Closing the wounds with sticky materials has been introduced to prevent bleeding and induce the wound healing process.
➢ Many types of surgical adhesives including tissue adhesives have been developed to be a suitable alternative for sutures and staples.
➢ A new approach which aims at producing bio-adhesives by mimicking the nature along with applying nanotechnology methods.
Self-healing adhesives
➢ Engineering stable underwater adhesives currently poses a major technical challenge, because most man-made adhesives fail in wet conditions, owing to chemical modification of the adhesive or its substrate.
➢ Diatoms produce adhesives that are extremely strong and robust, in both fresh- and seawater environments.
➢ AFM (Atomic Force Microscopy) study of living diatoms, phase imaging and force spectroscopy experiments revealed characteristics of these natural adhesives that may be of use in designing man-made adhesive analogues that function in wet environments.
➢ The AFM allows for investigations of micromechanical properties of the cell surface, for example viscoeleastic properties, adhesion forces and hardness measurements, in physiological conditions.
➢ Experiments with three diatom species that produce outstanding natural adhesives: *Eunotia sudetica*, *Navicula seminulum* and an unidentified species.

Mcqs

Dr. Richard Feynman, one of America's most notable physicists, 1918-1988.

Iron nanoparticles are 10 to 1 000 times more reactive than commonly used iron powders.

Scientists have created tools for the early diagnosis of pancreatic cancer by attaching a molecule that binds specifically to pancreatic cancer cells to iron oxide nanoparticles that are clearly visible under

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magnetic resonance imaging (MRI).
Nanorobots will have a diameter of about 0.5 to 3 micron

8.What is Carbon Nanotubes?

Carbon nanotubes have a variety of commercial uses, including making sports equipment stronger and lighter weight. For example, a tennis racket made with carbon nanotubes bends less during impact, and increases the force and accuracy of the delivery.
Nanoparticle-treated tennis balls can keep bouncing twice as long as standard tennis balls. Who found first nanotube?
1970: Morinobu Endo-- First carbon filaments of nanometer dimensions, as part of his PhD studies at the University of Orleans in France.
He grew carbon fibers about 7 nm in diameter using a vapor-growth technique. Filaments were not recognized as nanotubes and were not studied.
1991:Sumio Iijima-- NEC Laboratory in Tsukuba-- used high-resolution transmission electron microscopy to observe carbon nanotubes.
Carbon nanotube
➢ Carbon nanotubes are large molecules of pure carbon that are long and thin and shaped like tubes, about 1-3 nanometers (1 nm = 1 billionth of a meter) in diameter, and hundreds to thousands of nanometers long.
➢ As individual molecules, nanotubes are 100 times stronger-than-steel and one-sixth its weight.

9.What is copolymer?2

A copolymer is a macromolecule containing two or more types of monomers and a block copolymer comprises these basic units or monomer types joined together in long individual sequences called blocks.

10.Liquid crystals?

Liquid crystals
➢ A liquid crystal is a fourth state of matter: it has properties between those of a conventional liquid and those of a solid crystal.
➢ Like a liquid, it flows, and, like a crystal, it can display long-range molecular order.
➢ In terms of classifications, liquid crystals (together with polymers and colloids), are often classified as soft matter and treated under the branch of physical chemistry of condensed matter.

11.Diatoms are sever as model of biometric ?

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Self-healing adhesives

- Diatoms are a type of algae with nanostructured amorphous silica surfaces.
- Diatoms are unicellular algae that have the capability to colonize any natural and manmade submerged surfaces.
- Diatoms are a large group of eukaryotic microalgae that have a rigid cell wall made of silica (SiO_2).
- Diatoms are often among the first organisms to colonize underwater surfaces, because their adhesive mucilage enables the cells to attach to substrata of a very wide range of chemical composition (organic, inorganic, natural or man-made).
- Surface adhesion of diatoms belonging to the raphid pennate type requires the secretion of so-called adhesive mucilage strands through a dedicated slit in the silica cell wall, which is termed the raphe. Adhering cells can then move across the surface, which is accomplished through an actinmyosin dependent cytoskeletal motor that translocates the adhesive mucilage strands in a rearward direction through the raphe, thereby moving the cell forward.
- Some diatom species have evolved strong self-healing underwater adhesives. Some are free-floating, others have adhesive properties in water: for instance, diatoms in the Antarctic seas can attach to ice.
- Others secrete viscous mucilage which binds colonies together while protecting the silica shells from wear as they rub against each other.
- Molluscs are another species that have adhesive properties under water.
- Both diatoms and molluscs have strong underwater glues that can also resist stress and self-heal.
- They serve as a biomimetic model for self-healing materials

12.Difference between intramolecular and intermolecular forces?

Intramolecular forces are the forces that hold atoms together within a molecule. Intermolecular forces areforces that exist between molecules.

13.What is Nanotechnology?

Nanotechnology

- ❑ Simplest _ technology at the nonscale
- ➢ Nanotechnology literally means any technology on a nanoscale that has applications in the real world.
- ➢ Succinct definition of nanotechnology is engineering with atomic precision or atomically precise technology _ APT
- ➢ US National nanotechnology _ the essence of nanotechnology is the ability to work at the molecular level, atom-by-atom, to create large structure with fundamentally new molecular organization.
- ➢ Nanotechnology is group of emerging technology in which the structure of matter is

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controlled at the nanometer scale to produce novel materials and device that have useful and unique properties.

14.Environmental applications of Metal Nano Particles?

Metal nanoparticles _ MNPs

- The term metal nanoparticleis used to described nanosizedmetals with dimensions (length, width or thickness) within the size range 1-100 nm.
- The existence of metallic nanoparticlesin solution was first recognized by Faraday in 1857 and a quantitative explanation of their colourwas given by Mie in 1908.
- Metal nanoparticles are a clear example of how the properties of matter can change at the nanometre scale.
- For instance, metal gold is notably yellow in colour and used for jewellery. As the noblest of all metals, gold is very stable

Environmental applications

- Some forms of metal nanoparticles have important environment applications.
- Zero-valent (Fe0) iron nanoparticles are under investigation for the remediation of contaminated groundwater and soil.
- Iron, when exposed to air, oxidises easily to rust; however, when it oxidises around contaminants such as trichloroethylene (TCE), carbon tetrachloride, dioxins, or PCBs, these organic molecules are broken down into simple, far less toxic carbon compounds.
- Iron nanoparticles are more effective than conventional 'iron powder', which is already used to clean up industrial wastes. Iron nanoparticles are 10 to 1 000 times more reactive than commonly used iron powders.
- Silver nanoparticles have a strong antibacterial capacity.
- They are used in numerous products to prevent or reduce the adherence of bacteria to surfaces.

15.Applications of Carbon Nanotubes?

Carbon nanotube Applications

- CNTs have not only unique atomic arrangements but also have unique properties that include large current carrying capability

- long ballistic transport length
- high thermal conductivity
- mechanical strength

- These extraordinary properties of CNTs qualifies them exciting prospects and variety of

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applications in the area of microelectronics/nanoelectronics
- spintronics

optics
- material science
- mechanical

biological fields: Carbon nanotube Applications
➢ Particularly, in the area of nanoelectronics, CNTs and graphene nanoribbons (GNRs) demonstrates wide range of applications such as
-energy storage devices
-energy conversion devices that includes thermoelectric
-photovoltaic devices
-field emission displays and radiation sources
-nanometer semiconductor transistors
-nanoelectromechanical systems (NEMS
-electrostatic discharge (ESD) protection
-interconnects and passive
Carbon nanotube Applications - Structural
➢ CNTs possesses remarkable properties and qualities as structural materials. Their potential applications include:
(a) Textiles—CNTs can produce waterproof and tear-resistant fabrics.
(b) Body armor—CNT fibers are being used as combat jackets. The jackets are used to monitor the condition of the wearer and to provide protection from bullets.
(c) Concrete—CNTs in concrete increases its tensile strength and halt crack propagation.
(d) Polyethylene—CNT fibers can be used as polyethylene. The CNT based polyethylene can increase the elastic modulus of the polymers by 30 %.
(e) Sports equipment—Golf balls, golf clubs, stronger and lighter tennis rackets, bicycle parts, and baseball bats.
(f) Bridges—CNTs may be able to replace steel in suspension and bridges.
(g) Flywheels—The high strength/weight ratios of CNTs enable very high rotational speeds.
(h) Fire protection—Thin layers of buckypaper can potentially protect the object from fire. The dense, compact layer of CNT or carbon fibers in the form of buckypaper can efficiently reflect the heat.
Carbon nanotube Applications - Electromagnetic
➢ CNTs can be fabricated as electrical conductors, semiconductors and insulators. Such applications include:

Buckypaper—Thin nanotube sheets are 250 times stronger and 10 times lighter than steel. They can be used as heat sink for chipboards, backlight for LCD screens, or Faraday cage to protect electrical devices/aeroplanes.
(b) Light bulb filament—CNTs can be used as alternative to tungsten filaments in incandescent lamps

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(c) Magnets—A strong magnetic field can be generated using multi-walled CNTs coated with magnetite
(d) Solar cells—Germanium CNT diode exploits the photovoltaic effect. In some solar cells, nanotubes are used to replace the ITO (indium tin-oxide) to allow the light to pass to the active layers and generate photocurrent
(e) Electromagnetic antenna—CNTs can act as an antenna for radio and other electromagnetic devices due to its durability, light weight and conductive properties.

16.Enlist five application of Nanotechnology from.our daily life?

10 ways nanotechnology impacts our lives on a daily basis.
1. Faster, smaller, and more powerful computers that consume far less power, with longerlasting batteries
2. Faster, more functional, and more accurate medical diagnostic equipment. Lab-on-a-chip technology enables point-of-care testing in real time, which speeds up delivery of medical care. Nanomaterial surfaces on implants imp. Nanoparticles in pharmaceutical products improve their absorption within the body and
make them easier to deliver, often through combination medical devices. Nanoparticles can also be used to deliver chemotherapy drugs to specific cells, such as cancer cells.
4. Improved vehicle fuel efficiency and corrosion resistance by building vehicle parts from nanocomposite materials that are lighter, stronger, and more chemically resistant than metal. Nanofilters remove nearly all airborne particles from the air before it reaches the combustion chamber, further improving gas mileage.
5. Nanoparticles or nanofibers in fabrics can enhance stain resistance, water resistance, and flame resistance, without a significant increase in weight, thickness, or stiffness of the fabric. For example, “nano-whiskers” on pants make them resistant to water and stains.
6. Water filters that are only 15-20 nanometers wide can remove nano-sized particles, including virtually all viruses and bacteria. These cost-efficient, portable water treatment systems are ideal for improving the quality of drinking water in emerging countries.
7. Carbon nanotubes have a variety of commercial uses, including making sports equipment stronger and lighter weight. For example, a tennis racket made with carbon nanotubes bends less during impact, and increases the force and accuracy of the delivery. Nanoparticle-treated tennis balls can keep bouncing twice as long as standard tennis balls.
8. Most sunscreens today are made from nanoparticles that effectively absorb light, including the more dangerous ultraviolet range. They also spread more easily over the skin. These same nanoparticles are also used in food packaging to reduce UV exposure and prolong shelf life.
9. Many drink bottles are made from plastics containing nanoclays, which increase resistance to permeation by oxygen, carbon dioxide, and moisture. This helps retain carbonation and pressure and increases shelf life by several months.
10. Thanks to nanotechnology, a huge variety of chemical sensors can be programmed to

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detect a particular chemical at amazingly low levels, for example, a single molecule out of billions. This capability is ideal for surveillance and security systems at labs, industrial sites, and airports. On the medical front, nanosensors can also be used to accurately identify particular cells or substances in the body.

17.What is polymer nano-composite?

Although the scientific community has made remarkable progress in this field in the last years, polymer nanocomposites have just started to be explored, and many research questions still need to be addressed.

Polymer nanocomposites (PNC) consist of a **polymer** or copolymer having nanoparticles or nanofillers dispersed in the **polymer** matrix. ... **Polymer** nanoscience is the study and application of nanoscience to **polymer**-nanoparticle matrices, where nanoparticles are those with at least one dimension of less than 100 nm.

18.Write three applications of biotechnology?

Biotechnology – using living organisms, or the products of living organisms, for human benefit to make a product or solve a problem.
Biotechnology has been defined as the application of scientific and engineering principles to the processing of materials by biological agents to provide goods and services

Biotechnology is all around us and a big part of our lives, providing breakthrough products to cure disease, protect against bio-terrorism, feed the hungry, and clean our environment.
❑ Example of "modern" biotechnology
Recombinant DNA technology started modern biotech as an industry
❑ Examples of applications
- development of disease-resistant plants
- food crops that produce greater yields
- golden rice _ engineered to be more nutritious

19.Application of nanotechnology in the field of medicine?

Bionanotechnology is biological applications of nanotechnology (science and technology of miniaturization at scales of <100nm)
It is hoped that nanotechnology can deliver a valuable set of research tools and clinically helpful devices in the near future.
The Nanotechnology Initiative expects new commercial applications to be developed in the pharmaceutical industry including advanced drug delivery systems, new therapies, and in vivo imaging.
Neuro-electronic interfaces and other nanoelectronics-based sensors are also current goals of research.

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In the speculative field of molecular nanotechnology it is thought that cell repair machines could further revolutionize the field of medicine.

Nanotechnology is designed to provide a novel and improved approach to cancer diagnosis and treatment.
Nanoscale devices can interact with large biological molecules on both the surface and inside cells involved in cancer.
Since biological processes, including events that lead to the development of cancer, occur on a nanoscale at the surface of and inside cells, nanotechnology offers many tools.

Benefits for diagnosis
➢ In the fight against cancer, winning half the battle is based on early detection.
➢ Nanotechnology is contributing new molecular agents and methods to enable earlier and more accurate diagnoses and treatment monitoring.
❑ Imagine instead if cancerous or even precancerous cells could somehow be tagged for detection by conventional scanning devices. Two things would be necessary:
- Something that specifically identifies a cancerous cell and
- Something that enables it to be seen
➢ Antibodies that identify specific receptors found to be over-expressed in cancerous cells can be coated on to nanoparticles that then produce a high contrast signal when Magnetic Resonance Images (MRI) or Computed Tomography (CT) scans are used.
Nanotechnology imaging in cancer diagnosis
➢ Nanoparticles can enhance the efficacy of magnetic resonance imaging (MRI) in detecting the spread of cancer.
o In clinical trials, lymphotropic iron oxide nanoparticles acted as effective contrast agents and allowed the detection of small nodal metastases in men with prostate cancer that would otherwise have been overlooked.
o Nanoparticulate iron oxide particles were used with MRI to accurately detect metastatic lesions in lymph nodes without surgery.
➢ Nanoparticle contrast agents for ultrasound have also been developed that can enhance the sensitive detection of vascular and cardiac thrombi, as well as solid tumors of the colon, liver and breast, in a noninvasive manner.
Biomarker Screening
➢ Diagnostic screening for biomarkers in tissues and fluids could also be enhanced and potentially revolutionized by nanotechnology.
➢ Individual cancers differ from each other and from normal cells by changes in the expression and distribution of tens to hundreds of molecules.
➢ As therapeutics advance, it may require the simultaneous detection of several biomarkers may be required to identify a cancer for treatment selection

Nanoscale cantilevers and nanowire sensors
can detect biomarkers of cancer from a single
cell.

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➢ Nanoparticles such as quantum dots, which emit light of different colors depending on their size, could enable the simultaneous detection of multiple markers.

Nanotechnology_____ nanoscale

American Physicist name_____ Dr. Richard Feynman

By 2050, world population_____ 9.1billion

Human cell contain_____ 3 trillion

Gold properties wala-------

Diatom cell wall made of _____ silica (SiO2).

Nanotechnology not about _____ pm scale

Gold particles find significant exploitation_____ (i) their comparative chemical stability, making them less hazardous, (ii) simple and straightforward synthesis and fabrication process, and (iii) genuine biocompatibility and noninterference with other labeled biomaterials

Energy b/w valence n conduction band_____ Eg

Semiconductor have Eg_____ is between valence and conduction

Nanometer about_____10 H atom

Carbon tube width_____3nm

20.Nanotechnology ?

Ans: Simplest _ technology at the nonscale. Nanotechnology literally means any technology on a nanoscale that has applications in the real world. Succinct definition of nanotechnology is engineering with atomic precision or atomically precise technology _ APT. US National nanotechnology _ the essence of nanotechnology is the ability to work at the molecular level, atomby-atom, to create large structure with fundamentally new molecular organization. Nanotechnology is group of emerging technology in which the structure of matter is controlled at the nanometer scale to produce novel materials and device that have useful and unique properties

21.What are Self Assembled Monolayers ?

Ans: Self-assembled monolayers (SAMs)

Some organic molecules, when exposed from solution or vapour to a suitable substrate, selfassemble to produce homogeneous, densely packed layers of monomolecular thickness. These organic molecules have long chains with two different end groups. monolayer is formed when one of the two end groups

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of the organic molecule reacts with a particular surface forming a chemical bond. The surface properties of the substrate are then defined by the exposed functional groups of the monolayer.

For example, alkyl-silane or alkane thiol molecules exposed to a silica or metal surface assemble into organised layers.

22.Environmental application of Metal Nano Particles ?

Ans: Some forms of metal nanoparticles have important environment applications. Zero-valent (Fe0) iron nanoparticles are under investigation for the remediation of contaminated groundwater and soil. Iron, when exposed to air, oxidises easily to rust; however, when it oxidises around contaminants such as trichloroethylene (TCE), carbon tetrachloride, dioxins, or PCBs, these organic molecules are broken down into simple, far less toxic carbon compounds.Iron nanoparticles are more effective than conventional 'iron powder', which is already used to clean up industrial wastes. Iron nanoparticles are 10 to 1 000 times more reactive than commonly used iron powders. Silver nanoparticles have a strong antibacterial capacity. They are used in numerous products to prevent or reduce the adherence of bacteria to surfaces

23.Applications of Nanotechnology in diagnostics ?

Ans: Nanodiagnostics is an evolving application of nanoscale technology to meet the demand of clinical diagnostics, determining disease state, any predisposition to such, the pathology of the condition and the identification of causative organisms. With nanotechnology, diagnosis is being carried out on a nano-scale leading to a trend of the use of hand held devices that are easy to use and marketable. Nanodiagnostics as surging new field of molecular diagnostics, have been positively changing laboratory procedures, providing new ways for patient's sample assessment and early detection of disease biomarkers with increased sensitivity and specificity. Nanoparticle platforms have been developed and optimized for the detection of pathogens and cancer biomarkers such that diagnostic proceedures now become less cumbersome but more sensitive because most of the complex proceedures are now integrated onto a simple device having the capacity to be used for on the spot diagnosis

24.Applications of Nanotechnology in the field of Medicine?

Ans: Diagnostic Applications of Nanotechnology in Medicine

Improved imaging of the human (or any) body

Nanoprobes (miniature machines) can attach themselves to particles in the body (e.g., antibodies) and emit a magnetic field.

Probes that aren't attached to anything don't create a detectable magnetic

Nano-tracking may be able to detect tumors that are a few cells in size.

A single inhaled nanorobot reaches, deeply inspired into the lungs, enters an alveolar duct and attaches to the tissue surface.

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Another way to use nanotech as tracking devices is to use "quantum dots

These tiny semiconductors are able to emit wavelengths of light (colors) that depend on their size. If quantum dot A is twice as big as quantum dot B, it will emit a different color.

Quantum dots are better than conventional dyes:

They last much longer

More colors can be made available.

A microscopic machine roaming through the bloodstream, injecting or taking samples for identification and determining the concentrations of different compounds

Therapeutic Applications of Nanotechnology in Medicine

Nanotech is capable of delivering medication to the exact location where they are needed – hence lesser side effects

Organic dendrimers - a type of artificial molecule roughly the size of a protein- would be ideal for the job of delivering medicine

Hollow polymer capsules - gold-coated glass beads that are near infrared light sensitive

Destruction of harmful eukaryotic organisms / cancer cells by interrupting their division process

Certain proteins are capable of doing this (e.g., Bc12 family of proteins)

Nanoprobe can be made to generate radiation, that could kill bacteria, viruses and cancer cells

Nanoprobe comprising of a single caged actinium-225 atom would detect (using antibodies) and enter a cancerous cell

Location and destruction of cancer cells by acoustic signals

Comparison of normal and cancerous cells in respiratory airway of the lung

Nanotechnology also theoretically allows the mimicking of natural biological processes e.g., repair of damaged tissues

25.What is block copolymers (2)

Ans: A copolymer is a macromolecule containing two or more types of monomers and a block copolymer comprises these basic units or monomer types joined together in long individual sequences called blocks.

An example is the diblock polymer (A)m(B)n, which is made of a linear sequence of m monomers of type A joined together to a linear sequence of n monomers of type B.

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26.Current uses of nanotechnology. (3)

Ans: Nanotechnology is not an industry; nor is it a single technology or a single field of research The field of nanotechnology is interdisciplinary.

Nanotechnology is referred to as a general-purpose technology.

The total market for nanobiotechnology products was $19.3 billion in 2010 and is growing at a compound annual growth rate (CAGR) of 9% to reach a forecast market size of $29.7 billion by 2015.

Medical applications, including drug delivery and microbicides, dominate today's market, with sales of $19.1 billion in 2010. This market segment is growing at a compound annual growth rate (CAGR) of 8.7%, and is forecast to reach sales of $29 billion by 2015 (BCC). More and more consumer products are branded worldwide as nano.

27.Explain self-healing adhesive and why diatoms are biommitic models of self healing materials. (5)
Ans:

Diatoms are a type of algae with nanostructured amorphous silica surfaces.

Diatoms are unicellular algae that have the capability to colonize any natural and manmade submerged surfaces.

Diatoms are a large group of eukaryotic microalgae that have a rigid cell wall made of silica (SiO_2).

Some diatom species have evolved strong self-healing underwater adhesives. Some are free-floating, others have adhesive properties in water: for instance, diatoms in the Antarctic seas can attach to ice.

Others secrete viscous mucilage which binds colonies together while protecting the silica shells from wear as they rub against each other.

Molluscs are another species that have adhesive properties under water.

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They serve as a biomimetic model for self-healing materials.

These natural adhesives and their self-healing properties are due to the properties of the proteins contained in them.

These proteins have sacrificial bonds that allow the molecule to be reversibly stretched by breaking and re-bonding.

This sacrificial bond behaviour has been observed in many other materials, such as wool.

Closing the wounds with sticky materials has been introduced to prevent bleeding and induce the wound healing process.

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Many types of surgical adhesives including tissue adhesives have been developed to be a suitable alternative for sutures and staples.

A new approach which aims at producing bio-adhesives by mimicking the nature along with applying nanotechnology methods.

Engineering stable underwater adhesives currently poses a major technical challenge, because most man-made adhesives fail in wet conditions, owing to chemical modification of the adhesive or its substrate.

Diatoms produce adhesives that are extremely strong and robust, in both fresh- and seawater environments.

AFM (Atomic Force Microscopy) study of living diatoms, phase imaging and force spectroscopy experiments revealed characteristics of these natural adhesives that may be of use in designing man-made adhesive analogues that function in wet environments.

The AFM allows for investigations of micromechanical properties of the cell surface, for example viscoeleastic properties, adhesion forces and hardness measurements, in physiological conditions.

Experiments with three diatom species that produce outstanding natural adhesives: Eunotia sudetica, Navicula seminulum and an unidentified species

28.Three Applications of Biotechnology 2 marks

Ans: This reflects the current needs of humans- we have too many diseases and currently too few drugs to target them.

By far, cancer has the most drug candidates than any other disease. This disease has many more drug candidates than hepatitis C because hepatitis C affects less people worldwide than different kinds of cancer. Cancer, of course, can affect many different organs. On the other hand, hepatitis C only affects the liver.

29.Write the phases of crystal liquid 3 marks

Ans: Nematic Phase: Molecules in this phase are long and rod-like in shape. They are free to move in space.

Smectic Phase: This phase can be reached at lower temperatures than the nematic phase. Molecules align themselves in layers. More order and higher viscosity.

Cholestric Liquid Crystal: Molecules with intermolecular forces that favor alignment between molecules at a slight angle to one another

30.Application of SAM 3 marks

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Ans: Applications: SAMs are extremely useful in thin-film technology since they provide a simple, versatile and relatively inexpensive way of producing coatings with specific functionalities.

Thin coatings are used in many applications, for example to control surface wetting and adhesion properties; to provide chemical resistance or specific biocompatibility; to confer antibacterial properties; and for sensors.

SAMs are particularly interesting for developing DNA and protein micro and nanoarrays for medical screening and for future electronic components in computer chips and other ICT developments ways 31.nanotechnology impacts our lives daily basis?.) 5 marks

Ans: 1. Faster, smaller, and more powerful computers that consume far less power, with longerlasting batteries

Faster, more functional, and more accurate medical diagnostic equipment. Lab-on-a-chip technology enables point-of-care testing in real time, which speeds up delivery of medical care. Nanomaterial surfaces on implants improve wear and resist infection.

Nanoparticles in pharmaceutical products improve their absorption within the body and make them easier to deliver, often through combination medical devices. Nanoparticles can also be used to deliver chemotherapy drugs to specific cells, such as cancer cells.

Improved vehicle fuel efficiency and corrosion resistance by building vehicle parts from nanocomposite materials that are lighter, stronger, and more chemically resistant than metal. Nanofilters remove nearly all airborne particles from the air before it reaches the combustion chamber, further improving gas mileage.

Nanoparticles or nanofibers in fabrics can enhance stain resistance, water resistance, and flame resistance, without a significant increase in weight, thickness, or stiffness of the fabric. For example, “nano-whiskers” on pants make them resistant to water and stains.

Water filters that are only 15-20 nanometers wide can remove nano-sized particles, including virtually all viruses and bacteria. These cost-efficient, portable water treatment systems are ideal for improving the quality of drinking water in emerging countries.

Carbon nanotubes have a variety of commercial uses, including making sports equipment stronger and lighter weight. For example, a tennis racket made with carbon nanotubes bends less during impact, and increases the force and accuracy of the delivery. Nanoparticle-treated tennis balls can keep bouncing twice as long as standard tennis balls.

Most sunscreens today are made from nanoparticles that effectively absorb light, including the more dangerous ultraviolet range. They also spread more easily over the skin. These same nanoparticles are also used in food packaging to reduce UV exposure and prolong shelf life.

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Many drink bottles are made from plastics containing nanoclays, which increase resistance to permeation by oxygen, carbon dioxide, and moisture. This helps retain carbonation and pressure and increases shelf life by several months.

Thanks to nanotechnology, a huge variety of chemical sensors can be programmed to detect a particular chemical at amazingly low levels, for example, a single molecule out of billions. This capability is ideal for surveillance and security systems at labs, industrial sites, and airports. On the medical front, nanosensors can also be used to accurately identify particular cells or substances in the body

32.intermolecular and intramolecular bonding differwnce ?2 marks

Ans: Intramolecular bonding (chemical interactions): these are bondings that involve changes in the chemical structure of the molecules and include ionic, covalent and metallic bonds;

Intermolecular bonding (physical interaction): these are bondings that do not involve changes in the chemical structure of the molecules and include ion-ion and ion-dipole interactions; van der Waals interactions; hydrogen bonds; hydrophobic interactions; repulsive forces (such as steric repulsions).

33.quantum confinment 2 marks

Ans: The general principle is that confinement occurs if a characteristic size is less than or equal to the electron coherence length.

Quantum confinement causes the energy of the band gap to increase.

At very small dimensions when the energy levels are quantified, the band overlap present in metals disappears and is actually transformed into a band gap.

This explains why some metals become semiconductors as their size is decreased.

The increase in band gap energy due to quantum confinement means that more energy will be needed in order to be absorbed by the band gap of the material

34.biomimetic membrane application 3 marks

Ans: the important function of maintaining osmoregulation in living organisms by facilitating water transport through cell walls. Aquaporin molecule is the ability to restrict the passage of contaminants including bacteria, viruses, minerals, proteins, DNA, dissolved gases, salts, detergents, and even protons without encumbering the passage of water. For the past several years, a number of research efforts have been underway in an attempt to utilize aquaporins in the development of a new biomimetic membrane technology. Biomimetic membranes hold considerable promise as a low-energy option and a more effective alternative to desalination, industrial water treatment and other water purification applications. The Aquaporin Inside technology is essentially a thin film coating that hosts aquaporin proteins in an environment that retains the molecules natural activity of moving water molecules.

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